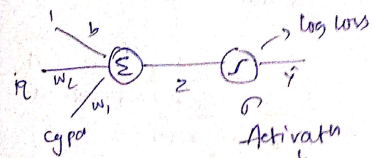


MLP Intuition

17/07/25



logistic

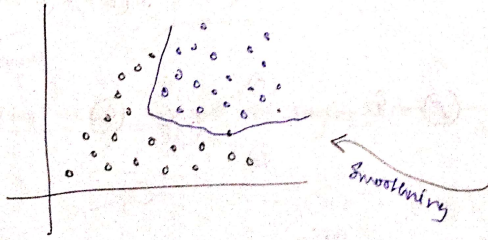
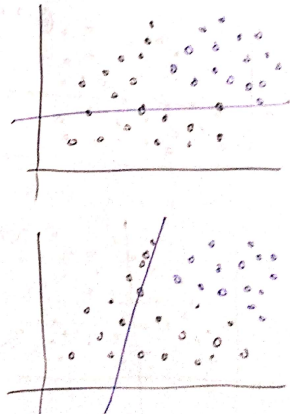
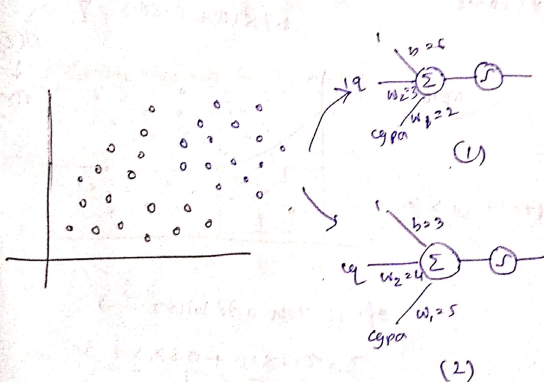
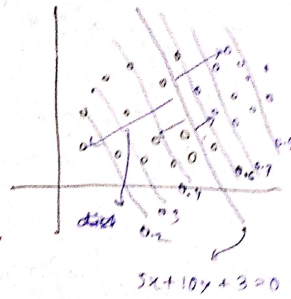
$w_1=5, w_2=10, b=3$

$\{8.7, 8.7\}$

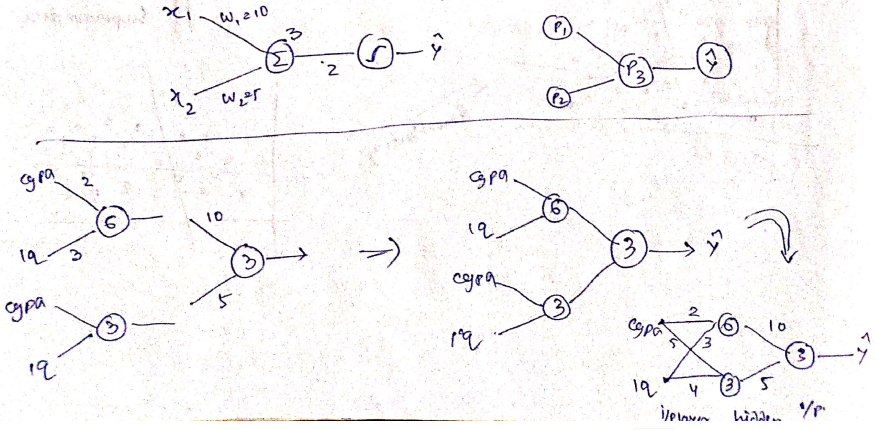
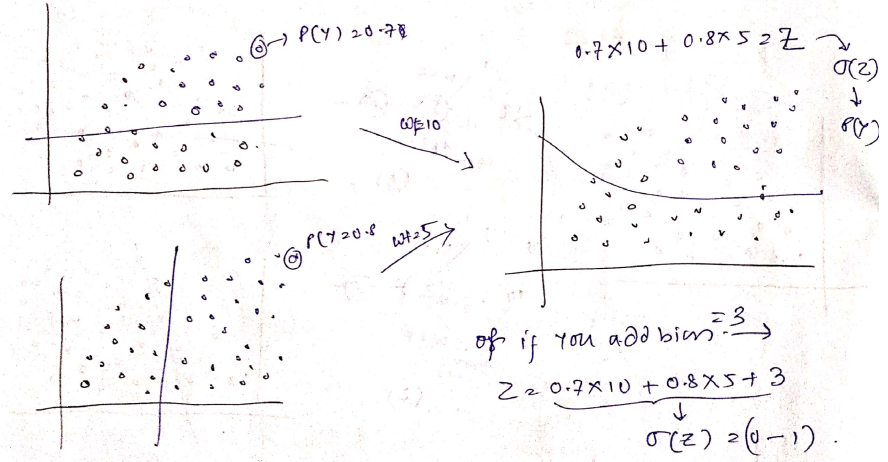
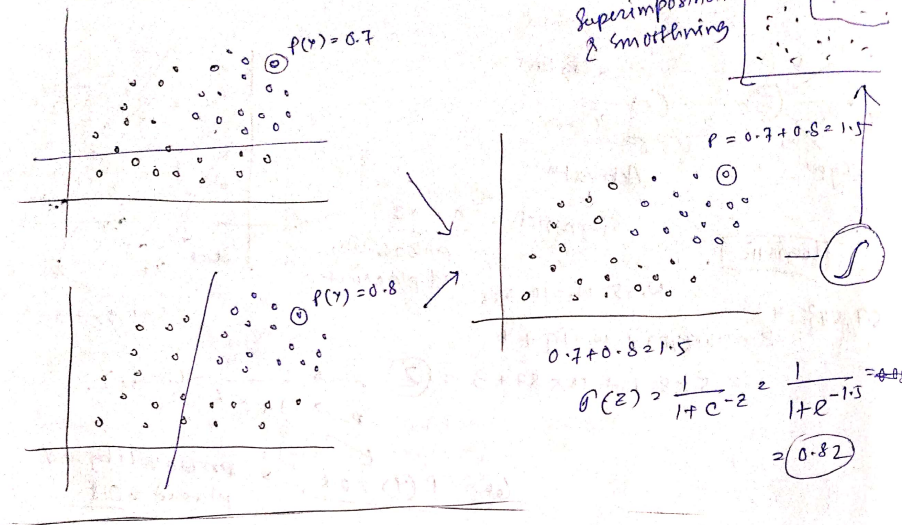
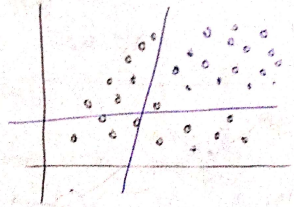
$$z = w_1 \cdot x_1 + w_2 \cdot x_2 + b$$

$$= 5 \times 8.7 + 10 \times 8.7 + 3 = 2 \rightarrow \frac{1}{1+e^{-z}} = (0 \rightarrow 1)$$

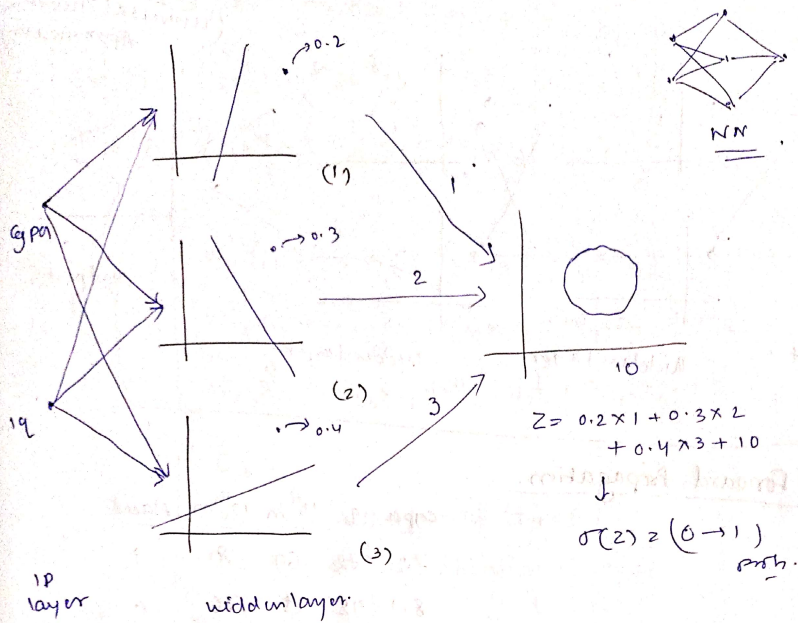
ex. $P(y) = 0.8 \rightarrow$ probability of placid ≈ 0.8



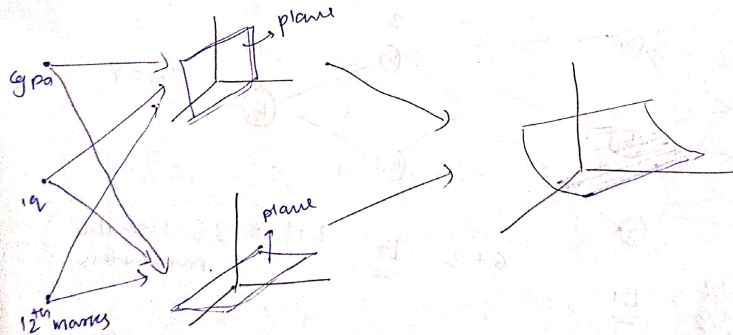
|| Superimpose



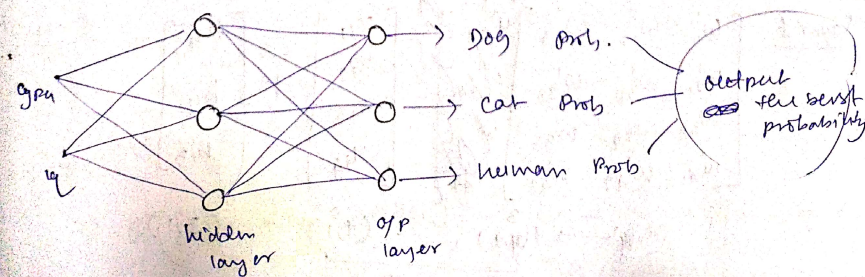
Adding nodes in hidden



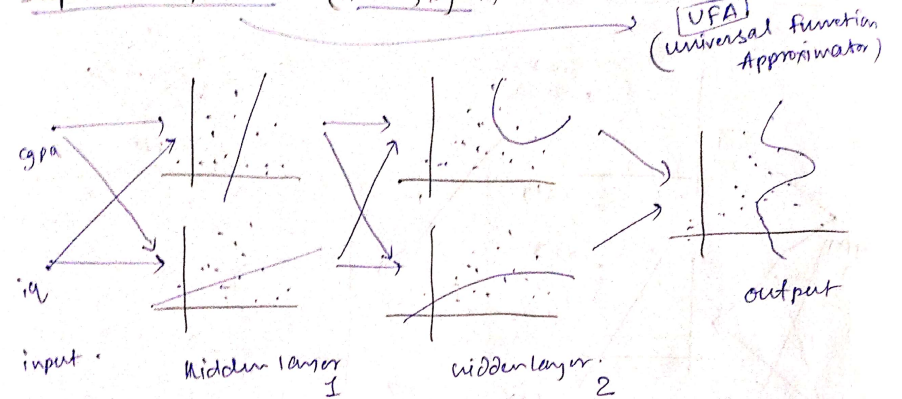
Adding Nodes in Input



Adding nodes in output layer node.

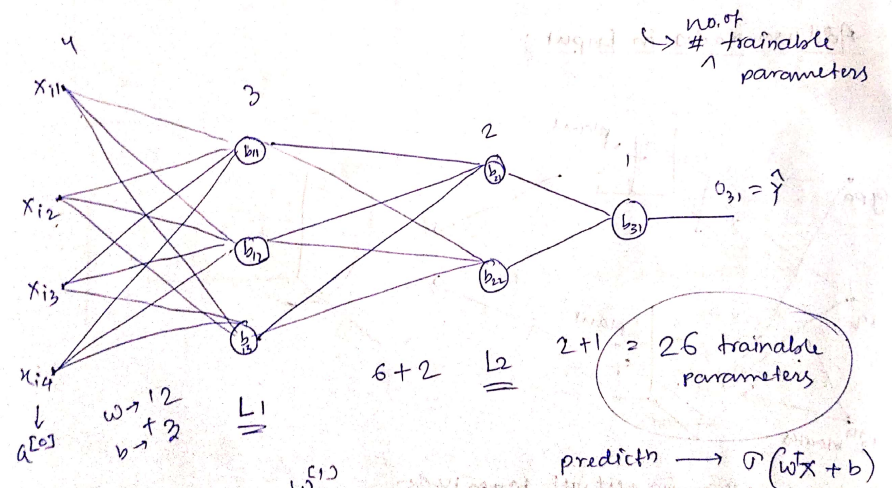


Deep Neural Network (Adding layers)



Forward Propagation.

cgpa	iq	10 th m	12 th m	placed
7.2	72	69	87	1
8.1	92	75	76	0



$$\text{layer 1} \begin{bmatrix} w'_{11} & w'_{12} & w'_{13} \\ w'_{21} & w'_{22} & w'_{23} \\ w'_{31} & w'_{32} & w'_{33} \\ w'_{41} & w'_{42} & w'_{43} \end{bmatrix} \begin{bmatrix} x_{11} \\ x_{12} \\ x_{13} \\ x_{14} \end{bmatrix} + \begin{bmatrix} b_{11} \\ b_{12} \\ b_{13} \end{bmatrix}$$

Diagram illustrating the forward propagation calculation for layer 1. The input vector X is multiplied by the weight matrix W and the bias vector b is added to the result. The output is then passed through a sigmoid function σ .